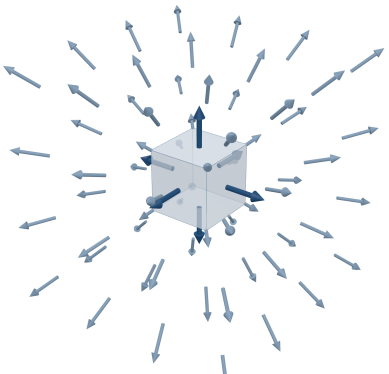
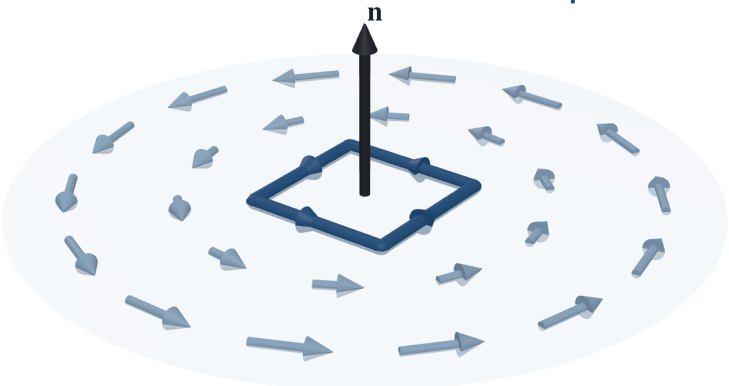
Divergence: net outflow of a small box



$$\oint \mathbf{F} \cdot \mathbf{n}\, \mathrm{d}S > 0 \;\Rightarrow\; \nabla \cdot \mathbf{F} > 0$$

Curl: circulation round a small loop



$$\oint \mathbf{F} \cdot \mathrm{d}\boldsymbol{\ell} > 0 \;\Rightarrow\; (\nabla \times \mathbf{F}) \cdot \mathbf{n} > 0$$